MIL-P-46490F(MU)
10 July 1974
SUPERSEDING
MIL-P-46490E(MU)
10 February 1965

MILITARY SPECIFICATION

PROJECTILE, HEAT-FS, 90MM, M431E1, METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Projectile, HEAT-FS, 90MM, M431E1 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-M-6867	- Magnetic Inspection Units
MIL-I-6868	- Inspection Process, Magnetic Particle
MIL-C-60536	- Coating, Anodic, Hard for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-410	- Certification of Magnetic Particle Inspection Personnel
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition

FSC 1315

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2.1 (Cont'd)

MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Armament Command

F8594965	- Metal Parts Assembly
LI-8594965	- Index Listing for Inspection Equipment for Projectile, HEAT-FS, 90MM, M431E1, Metal Parts Assembly
F7548604	- Pallet for Fixed and Semifixed Shell Assembly, Details, Packing and Marking

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1	- Surface Texture, Surface Roughness, Waviness and Lay

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

ASTM Method E-8	- Tension Testing of Metallic Materials
ASTM Method E-10	- Brinnell Hardness of Metallic Materials
ASTM Method B117	- Salt Spray (FOG) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

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3.2 Heat treatment or stress relief.-The heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in any component the mechanical properties specified on the applicable component drawing.

3.3 Wire assembly.

3.3.1 Pull test.-The wire and eyelet assembly shall withstand a pull test of 90 pounds, minimum.

3.3.2 Fuze wire terminal.-The fuze wire terminal when crimped to the fuze wire shall withstand a pull test of 20 pounds, minimum.

3.3.3 Insulation breakdown.-The wire assembly when assembled to the metal parts assembly, shall exhibit no evidence of electrical breakdown when subjected to a direct current potential differential of 1500 volts.

3.3.4 Electrical continuity.-The wire assembly when assembled to the metal parts assembly shall exhibit no evidence of electrical discontinuity.

3.3.5 Electrical resistance.-After assembly, the resistance between the forward end of the spike and the rear of the body shall not exceed 0.2 ohm, max.

3.4 Cone.

3.4.1 Spin compensation.-Cones shall spin compensated for a spin rate of 20 RPS $\pm$ 5 RPS. The process utilized for manufactures of the cones shall be such that the resulting grain orientation pattern will compensate for the clockwise rotation of the projectile.

3.4.2 Static firing penetration.-Cones shall be qualified by static firing the first article sample. The loaded body, spike, chamber and cover assembly, when fired at 20 RPS, shall penetrate not less than 14 inches of mild steel. Average penetration of all assemblies fired statically at 20 RPS shall be not less than 15 inches.

3.5 Protective coating.-Protective coating shall be in accordance with the applicable drawings and specifications. When protective coating is required by component details, finish requirements will apply to all indicated surfaces of the component regardless of contour configuration.

3.6 Painting.-Painting shall comply with the requirements on the applicable drawings. All paint and marking shall be dry to the touch before testing and packing for shipment.

3.7 Air pressure.-The explosive charge cavity of the assembly shall be capable of containing an internal air pressure of 3 $\pm$ .5 psi without evidence of leakage, for a minimum of 30 seconds.

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3.8 Metal defects.

3.8.1 Components.-All components, and component cavities, shall be free from cracks, splits, cold shuts, inclusions, pipe, porosity, bursts, folds, seams and other metal defects.

3.8.2 Component cavities.-In addition to 3.8.1, all component cavities shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.8.3 Visual standards.-Visual standards for metal defects will be established for each contract after the first three month's production. However, the establishment of visual standards will not waive the intent of the requirement for compliance with 3.8.1 and 3.8.2. (See 6.2)

3.9 Proving ground functional tests.-The projectile metal parts shall reveal no evidence of metal parts failure when tested at excess and service pressure. Metal parts failure shall be:

a. Loss or breakup of metal parts components except obturating band in the gun bore or in flight.

b. Penetration (The loaded projectile shall be capable of defeating six (6) inches of homogeneous armor at sixty (60) degrees obliquity at all useable ranges ninety percent of the time. Absolute penetration at all angles of obliquity up to sixty degrees shall be not less than twelve inches.)

3.10 Workmanship.-The requirements for workmanship are as specified on the applicable drawing, referenced specifications and the following:

3.10.1 Material defects.-Molded components shall be free of porosity and excessive flash.

3.10.2 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing.

3.10.3 Threads.-Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings.

3.10.4 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.10.5 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter prior to inspection and shipping.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspection required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground functional tests.

b. Acceptance of a lot by the Government shall be contingent on successful passage of the Proving Ground Functional tests.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 metal parts assemblies as follows:

35 metal parts assemblies, complete, including paint
5 metal parts assemblies, complete except paint
5 sets of sub-assemblies
5 sets of component parts unpainted, unassembled

Thirty of the above complete and painted metal parts assemblies shall be forwarded to a designated Government facility for loading and subsequent ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

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4.2.1.1 Cones.-Prior to the start of regular production and in order to establish form, size, temper, speeds and feeds of the flo-turn operation, a first article sample consisting of thirty-five (35) cones from each flo-turn machine in operation, without drill holes shall be assembled by the contractor into body, spike, cone, locking ring, fuze lock plug and chamber assemblies and forwarded to a designated Government facility for loading and subsequent static firing tests to determine compliance with 3.4.1 and 3.4.2. The contractor shall record and submit to the technical agnecy all information relative to the manufacture of the First Article cone sample, i.e., form, size, temper, speeds and feeds, etc. per each machine in operation, for subsequent use in the production of cones, after static firing results have been found acceptable. Failure of the static firing tests to comply with the requirements of 3.4.1 and 3.4.2 shall be cause for rejection of the first article sample. When the parameters to manufacture the cones have been established as a result of the successful static firing tests, the contractor shall maintain these processes and procedures throughout the contract while manufacturing the cones. Should any change in process, procedure or material occur while in production, a new first article sample shall be submitted as outlined in this paragraph.

4.2.1.2 Examination and test.-After inspection and provisional acceptance at source, the sample will be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.2.1.4 Abrasion resistance.

4.2.1.4.1 Test panels.-When the protective coating is being applied to the initial production sample of the Fin Assembly (or components thereof), 4 test panels as specified in paragraph 9.4.3 of MIL-C-60536(MU) shall be processed along with the product, through each step in the coating process. After coating the 4 panels will be forwarded to a Government approved laboratory for testing in accordance with 9.4.3 of MIL-C-60536(MU) to determine compliance with paragraph 8.4.1 of MIL-C-60536(MU).

4.2.1.4.2 File test.-At the same time the test panels provided by 4.2.1.4.1 are being subjected to the abrasion test at the Government approved laboratory 2 fin assemblies (or two sets of component parts thereof) from the initial production sample shall be subjected to the file test in accordance with paragraph 9.4.4 of MIL-C-60536(MU) to establish correlation (of hardness of coat) between the panels from 4.2.1.4 and the file test of the fin assemblies or components thereof. The file test will be performed using the equipment and procedures that will be used in production testing.

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4.2.1.4.3 Correlation.-When performing the file test prescribed by 4.2.1.4.2 the following information shall be obtained:

a. The type and model of the filing machine used.
b. The type of file (roughness) used.
c. The length of the file stroke.
d. The pressure applied to the file.
e. The number of strokes of the file required to expose the basic metal.

The information thus obtained shall be the basis for determining compliance of the production sample herein after prescribed, wth the requirements of paragraph 8.4.2 MIL-C-60536(MU).

4.3 Inspection provisions, test methods and procedures. (See 6.3)

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Mechanical properties.

4.3.1.2.1 Hardness and tension test.

4.3.1.2.1.1 Heat treatment lots (definition).-Heat treatment lots shall contain spikes and bodies from not more than one heat of steel, that has been heat treated in the same heat-treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.2.1.2 Heat treatment lots (Hardness).-In order to determine uniformity of heat treatment, each body and spike in the heat treatment lot (as defined above) shall be subjected to a hardness test with a Brinnel Hardness Tester, using a 5 second minimum dwell or by any other method of hardness testing approved by the procuring activity.

4.3.1.2.1.3 Heat treatment lots (Tension test sample).-Two bodies and two spikes (as applicable) shall be selected from each heat treatment lot, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of one or more tension test specimens to comply shall be cause for rejection of the lot of components represented by the failing specimen. On tangentially transverse specimens the elongation shall be recorded for information only. At the option of the supplier a

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4.3.1.2.1.3 (Cont'd)

production control may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Bodies or spikes which fall outside of the established range may be re-heat treated to fall within the range, or mechanical properties may be determined from newly selected samples which will represent the true extremes of hardness in the heat treatment lot.

4.3.1.2.1.4 Stress relieved lots (definition).-Stress relieved lots shall contain spikes and bodies cold shaped by the same process, utilizing the same cold shaping equipment (as applicable) and stress relieved in one unchanged process. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.2.1.5 Stress relieved lots (Hardness).-Twenty-five cold shaped bodies and spikes (as applicable) from the first batch of each stress relieved lot (as defined above) processed through the stress relief furnace and five bodies or spikes from each 1 hour production thereafter for the balance of the stress relieved lot shall be subjected to a hardness test to assess uniformity of the product.

4.3.1.2.1.6 Stress relieved lots (Tension test samples).-Two bodies and two spikes (as applicable) shall be selected from each stress relieved lot, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five bodies and spikes shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. On tangentially transverse specimens the elongation shall be recorded for information only. When it has been determined that the mechanical properties comply and the furnace temperature and time has been established, the furnace temperature and time shall be maintained throughout each stress relief lot. The furnace temperature may vary within twenty-five degrees fahrenheit but not below the minimum temperature of 700 degrees fahrenheit and the minimum 30 minutes soak time. Furance temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated each 8 hours by an independent method.

4.3.1.3.1.7 Miscellaneous lots (definition).-Small lots (such as spikes or bodies manufactured from abnormally small heats of steel or spikes or bodies that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with 4.3.1.2.1.1 or 4.3.1.2.1.4 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

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4.3.1.2.1.8 Miscellaneous lots (Hardness).-Miscellaneous heat treatment lots of spikes or bodies shall be hardness tested in accordance with 4.3.1.2.1.2. Miscellaneous cold shaping lots of bodies shall be hardness tested in accordance with 4.3.1.2.1.5.

4.3.1.2.1.9 Miscellaneous lots (Tension test sample).-Samples for tension test of miscellaneous heat treatment or stress relieved lot of bodies or spikes shall be selected and tested in accordance with 4.3.1.2.1.3 or 4.3.1.2.1.6 as applicable. Failure of one or more tension test specimens to comply with mechanical properties shall be cause for rejection of the lot of components represented by the failing specimen. On tangentially transverse specimens, the elongation shall be information only. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.3.1.2.1.10 Hardness test method.-After final heat treatment or stress relief, but prior to finish machining, one Brinnel hardness test shall be performed on the large outside diameter of the spike and on the rear face of the body. Prior to performing the test, the area in which the readings are to be taken shall be cleaned of scale or other foreign matter. The test shall be performed using equipment and procedures prescribed by ASTM Method E-10.

4.3.1.2.1.11 Tension test methods.

4.3.1.2.1.11.1 Body.-From the samples provided in accordance with the heat treat, stress relieved or miscellaneous lots, two tension test specimens conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the body wall. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and with the approval of the procuring agency, the Government shall not be restricted from testing additional longitudinal specimens at other locations to determine that the mechanical properties of the body meet the drawing requirements. The criteria for performing this test shall be prescribed by ASTM Method E-8, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.3.1.2.1.11.2 Spike.-From the samples provided in accordance with the heat treat, stress relieved or miscellaneous lots, two tension test specimens conforming to the largest obtainable standard "T" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the stem portion of the spike. In addition, two tension test specimens conforming to the largest obtainable standard "R" type as prescribed by ASTM Method E-8 shall be taken in a tangentially transverse direction from opposite sides at

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4.3.1.2.1.11.2 (Cont'd)

the midpoint of the flange area. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and with the approval of the procuring agency, the Government shall not be restricted from testing additional specimens at other locations to determine that the mechanical properties of the spike meet the drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.3.1.2.1.12 Fins (Tension test samples).-Two fins from each lot of 1200 fins or from 400 feet of extrusion or fraction thereof shall be subjected to the tension test. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot represented by the sample.

4.3.1.2.1.13 Fin (Tension test method).-From the samples proveded in accordance with 4.3.1.2.1.12, one tension test specimen conforming to the largest obtainable standard "R" type specimen shall be taken longitudinally from each component. The criteria for performing this test shall be as prescribed by ASTM Method E-8.

4.3.1.2.1.14 Boom (Tension test samples).-Two booms from each lot of 1200 or fraction thereof shall be subjected to the tension test. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot represented by the sample.

4.3.1.2.1.15 Boom (Tension test method).-From the samples provided in accordance with 4.3.1.2.1.14, one tension test specimen conforming to the largest obtainable standard "R" type specimen shall be taken longitudinally from each component. The criteria for performing this test shall be as prescribed by ASTM Method E-8.

4.3.1.2.1.16 Cone lots (definition).-A cone lot shall consist of cones produced with the same material, by one manufacturer in one unchanged process and procedure, in accordance with the same drawings, same drawing revisions, same specification and same specification revisions.

4.3.1.2.1.17 Cone (inspection).-Once the manufacturing processes and procedures have been established, as determined by 4.2.1.1 the established processes and procedures shall be maintained, controlled and monitored by the contractor on an hourly basis or fraction thereof, on each flo-turn machine in operation. Verification of this inspection shall be performed by the Government. Any change in a process, procedure or material shall be cause for rejection of the lot represented by the sample. All information relative to manufacture of cones will be recorded. The records will be made available to the Government upon request.

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4.3.1.2.2 Magnetic particle inspection.-After heat treatment or stress relief and finish machining, each body shall be subjected to a magnetic particle inspection. Failure of any body to comply with 3.8 shall be cause for rejection of the body.

4.3.1.2.3 Magnetic particle (Inspection method).-The procedures for performing this test utilizing the wet fluorescent method shall be as specified in MIL-I-6868, for indirect magnetization by either the continuous method or the residual method, utilizing a circular magnetic field. The suspension shall be applied to the entire body, inside and outside. After draining, the body shall be throughly inspected in a darkened area under an ultra-violet light which will produce adequate fluorescence of the magnetic particles. The solid content of the suspension shall be maintained at a level recommended by the suspension manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by specification MIL-I-6868. When performing the test for solid content of the suspension, if it is determined that it does not comply, all projectile bodies that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. Magnetic particle inspection shall be performed on equipment that complies with the requirements of MIL-M-6867. The operator(s) of the equipment and inspector(s) making the determinations shall be certified in accordance with MIL-STD-410.

4.3.1.2.4 Torque test samples.-Five fin and boom assemblies and five projectile sub-assemblies from each one hour's production or fraction thereof from each torque machine in operation shall be selected at random and tested to determine compliance with the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the production represented by the sample.

4.3.1.2.5 Fin and boom torque test method.-The fin and boom assemblies supplied in accordance with 4.3.1.2.4 shall be mounted in an approved holding device in such a manner that the boom is prevented from turning. The required torque shall be applied to the fin gradually in the direction required for assembly. Any perceptible movement of the fin shall be cause for rejection. Testing shall be conducted while thread sealant is wet.

4.3.1.2.6 Projectile sub-assembly torque test method (spike to body) and (chamber to body).-The projectile sub-assemblies supplied in accordance with 4.3.1.2.4 shall be mounted in an approved holding device in such a manner that the body is prevented from turning. The torque shall be applied to the spike or chamber (as applicable) in the direction of the assembly. Any perceptible movement of the spike or chamber (as applicable) shall be cause for rejection. Testing shall be conducted while thread sealant is wet.

4.3.1.2.7 Strength of wire assembly samples.-Ten wire and insulator assemblies from each lot of 5000 or fraction thereof shall be subjected to a pull test to determine compliance with 3.3.1. Failure of one or more assemblies to comply shall be cause for rejection of the lot represented by the sample.

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4.3.1.2.8 Strength of wire assembly method.-The samples selected in accordance with 4.3.1.2.7 shall be placed in an approved holding fixture and the load applied gradually in the direction of the wire axis. Assemblies tested shall not be used in metal parts assemblies.

4.3.1.2.9 Air pressure test.-Each metal parts assembly consisting of body lock ring, conduit and cone, components as a minimum shall be subjected to an air pressure test to determine compliance with 3.7. Failure of any assembly to comply shall be cause for rejection of that assembly.

4.3.1.2.10 Air pressure method.-The assembly shall be mounted in an approved holding fixture equipped with a gage to visibly determine the pressure applied and held for the minimum time required. A drop in pressure reading prior to the expiration of the required time will indicate a leaking assembly.

4.3.1.2.11 Insulation breakdown test.-Each metal parts assembly shall be subjected to this test to determine compliance with 3.3.3. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.1.2.12 Insulation breakdown test method.-The direct current shall be applied between the fuze wire terminal and the rear of the body.

4.3.1.2.13 Electrical continuity test.-Each metal parts assembly shall be subjected to this test to determine compliance with 3.3.4. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.1.2.14 Electrical continuity test method.-The voltage shall be applied between the eyelet terminal and the fuze wire terminal and the circuit monitored for evidence of discontinuity.

4.3.1.2.15 Electrical resistance test.-Each metal parts assembly shall be subjected to this test to determine compliance with 3.3.5. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.1.2.16 Electrical resistance test method.-The resistance between the forward end of the spike and the rear of the body shall be checked with an ohmeter capable of performing this test accurately and consistently.

4.3.1.2.17 Fuze wire terminal test.-The fuze wire terminals of 25 metal parts assemblies from each 8 hours production or fraction thereof shall be subjected to a pull test to determine compliance with 3.3.2. Failure of any terminal to comply shall be cause for rejection of the lot of metal parts assemblies represented by the sample.

4.3.1.2.18 Fuze wire terminal test method.-The samples selected in accordance with 4.3.1.2.17 shall be tested using an approved holding fixture and the load applied gradually to the direction of the wire axis.

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4.3.1.2.19 Protective coating.

4.3.1.2.19.1 Metal parts assembly sample.-The sample size and acceptance criteria for the inspection of cleaning the metal parts assembly and the application of the protective coating shall be in accordance with the requirements of the applicable drawing and the specifications referenced thereon. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.3.1.2.19.2 Metal parts assembly test method.-The procedures for testing the cleaning of assemblies and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specification listed thereon. The salt spray test shall be performed utilizing equipment prescribed by ASTM Method B-117.

4.3.1.2.19.3 Abrasion resistance test sample.-One fin assembly (or set of components thereof) from each 8 hours production or fraction thereof from the protective coating process shall be subjected to this test to determine compliance with paragraph 9.4.4 of MIL-C-60536(MU), except that the information obtained during file test in 4.2.1.4.3 shall be utilized as indicated in 4.2.1.4.3. Failure of the assembly or components thereof to comply shall be cause for rejection of the product processed since the last acceptable test was performed.

4.3.1.2.19.4 Abrasion resistance file test method.-The samples provided by 4.3.1.2.19.3 shall be tested using a filing machine of a type and model that was used in establishing correlation as indicated in 4.2.1.4.3. The samples shall be tested using the same type (roughness) file, the same length of stroke, the same number of strokes and the same file pressure as was used on the initial production sample.

4.3.1.3 Proving Ground functional tests.-The following Proving Ground tests will be performed by the Government:

4.3.1.3.1 First article sample.

4.3.1.3.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Phase	Quantity
Excess pressure	10
Service pressure	10
Service pressure (Penetration)	10

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4.3.1.3.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Phase	Quantity
Excess pressure	10 each lot
*Service pressure	10 each lot

* After five (5) consecutive lots have been ballistically tested and accepted, ballistic testing may be discontinued, with the approval of the procuring agency. However, implementation of ballistic testing will be instituted whenever directed by the procuring contracting officer.

4.3.1.3.3 Excess pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.8 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.1.3.4 Service pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.8 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.1.3.5 Excess pressure method.-The metal parts assemblies supplied in accordance with Table I and II for these phases shall be inert loaded to the service weight and assembled into complete rounds and fired in a gun for which the round is standard. The propellant charge will be adjusted to obtain 109 to 117 percent of the upper pressure limit of the propellant used with the applicable projectile. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.8.

4.3.1.3.6 Service pressure method.-The metal parts assemblies supplied in accordance with Tables I and II for this phase shall be inert loaded to the service weight and assembled into complete rounds and fired in a gun for which the round is standard. The rounds will be fired to impact on a vertical target placed 2000 meters from the gun muzzle. The test rounds will be fired alternately with the projectiles from a referenced lot for comparison of velocity-accuracy at this range. Accuracy in terms of maximum spread in inches, both horizontal and vertical and probable error (PE) in mils along with velocity and pressure of each round fired will be recorded for information purposes only. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.8.

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4.3.1.3.7 Penetration phase.-The metal parts assemblies supplied in accordance with Table I for this phase shall be loaded to the service weight and assembled into complete rounds and fired in a gun for which the round is standard. Against a six (6) inch thick homogeneous armor plate placed approximately 400 feet from the gun muzzle at an obliquity of 60 degrees.

4.3.1.3.8 Static firing samples (first article sample).-The thirty five (35) cones assembled in accordance with 4.2.1.1 shall be subjected to this test to determine compliance with 3.4.1 and 3.4.2. Failure of the penetration versus the rotation curve (plotted from the resulting data) to comply with the requirements of 3.4.2 shall be cause for rejection of the first article sample.

4.3.1.3.9 Static firing samples (production lot).-Ten (10) cones assembled in accordance with 4.2.1.1 from each production lot shall be subjected to this test to determine compliance with 3.4.2. Failure to comply with the requirements of 3.4.2 shall be cause for rejection of the lot represented by the sample. In the event of a test failure, ten additional samples shall be statically fired. Five (5) cones will be fired at 5 RPS above and five (5) cones will be fired at 5 RPS below the spin rate of the first test to assist in the correction of the spin compensation parameters. After five (5) consecutive lots have been statically fired and accepted, static firing tests may be discontinued, with the approval of the procuring agency. However, implementation of the static firing test will be instituted whenever directed by the procuring contracting officer.

4.3.1.3.10 Static firing test method (first article sample).-The samples provided in accordance with 4.3.1.3.8 will be loaded with composition "B" explosive and statically fired against a stacked target composed of mild steel blocks, 3 inches to 5 inches thick (100 to 120 BHN). These blocks will be used to build up the target to a thickness of 18 inches minimum. The standoff will be 7.05 inches. Standoff is defined as the distance between the cone base and the plate target. Five of the assemblies will be fired at the required 20 RPS frequency and the remaining 30 assemblies will be fired in groups of five (5) at 5 - 10 and 15 revolutions per second (RPS) above and below the 20 RPS frequency. A penetration in inches versus rotation in RPS curve will be prepared from the resulting data, plot all points. The curve will be defined by the mean penetration values at each frequency used.

4.3.1.3.11 Static firing test method (production lot).-The samples provided in accordance with 4.3.1.3.9 will be loaded with composition "B" explosive and statically fired against a stacked target composed of mild steel blocks 3 inches to 5 inches thick (100 to 120 BHN). These blocks will be used to build up the target to a thickness of 18 inches minimum. The standoff will be 7.05 inches. Standoff is defined as the distance between the cone base and the plate target. The assemblies will be fired at the required 20 RPS frequency.

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4.3.1.4 Inspection equipment.-Index of Inspection Equipment List Number LI-8594965 identified the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation specified on the applicable LI.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity (See 6.4). An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Tool control dimensions.-Dimensions marked tool control shall be gaged during First Article testing and after each 250 items produced on each machine in operation and after any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.3.2.2 Spike, Drawing F8594971, covering a detail of Drawing F8594965.

Defect	Method of Inspection
CRITICAL	
1. Braze metal not visible for full 360 degrees (if applicable)	Visual
2. Metal defective	Visual
MAJOR	
101. Runout of bell housing with pitch diameter of large thread	Gage
102. True position of insulator cavity with pitch diameter of small thread	Gage
103. Pitch diameter of large thread, min.	Gage
104. Major diameter of large thread, min.	Gage
105. Diameter of spike stem (inside)	Gage
106. Diameter of spike stem outside (small)	Gage
107. Diameter of spike stem outside (medium)	Gage

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
108.	Diameter of spike stem outside (large)	Gage
109.	Diameter of small thread undercut	Gage
110.	Diameter at rear of bell housing	Gage
111.	Diameter of insulator cavity	Gage
112.	Diameter rear of insulator cavity	Gage
113.	Diameter of wire hole (if applicable)	Gage
114.	Diameter of brazing surfaces (if applicable)	Gage
115.	Depth of insulator cavity	Gage
116.	Distance to bottom of wire slot	Gage
117.	Length from front face to medium outside diameter of spike stem	Gage
118.	Length from front face of spike flange to large outside diameter of spike stem	Gage
119.	Length from front face to nose cap seat	Gage
120.	Length to basic diameter (bell housing)	Gage
121.	Length to basic diameter rear of spike flange	Gage
122.	Length of effective large thread, min.	Gage
123.	Length of brazing surface (if applicable)	Gage
124.	Length of adapter (if applicable)	Gage
125.	Length, total	Gage
126.	Length from face of adapter to rear of flange (if applicable	Gage
127.	Location of wire hole (if applicable)	Gage
128.	Variation in wall thickness of spike stem	Gage
129.	Width of wire slot	Gage
130.	Misalignment of adapter to spike stem, max.(if applicable)	Gage
MINOR		
201.	Diameter of large thread undercut, min. (if applicable)	Gage
202.	Diameter of undercut rear of adapter flange (if applicable)	Gage
203.	Diameter rear of spike flange	Gage
204.	Angle rear of spike flange	Gage
205.	Length from rear face to front of flange	Gage
206.	Length from rear face to rear of flange	Gage
207.	Width of small thread undercut	Gage
208.	Width of large thread undercut, max. (if applicable)	Gage
209.	Width of undercut rear of adapter flange (if applicable)	Gage
210.	Radii or chamfer missing or incorrect	Visual (See 6.1)
211.	Surface finish improper	Visual (See 6.1)
212.	Burr	Visual
213.	Foreign matter	Visual

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4.3.2.3 Cone, Drawing D8594975, covering a detail of Drawing F8594965.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Diameter of flange	Gage
102. Diameter inside (rear)	Gage
103. Diameter outside (rear)	Gage
104. Diameter of conduit hole	Gage
105. Diameter rear of flange	Gage
106. Length to forward basic diameter	Gage
107. Length to rear basic diameter	Gage
108. Length, total	Gage
109. Location of conduit hole	Gage
110. Thickness of wall	Gage
111. Thickness of wall (rear face)	Gage
112. Thickness of flange	Gage
113. Variation in wall thickness circumferential	Gage
114. Variation in slant length	Gage
MINOR	
201. Radii or chamfer missing or incorrect	Visual (See 6.1)
202. Surface finish improper	Visual (See 6.1)
203. Burr	Visual
204. Foreign matter	Visual

4.3.2.4 Body, Drawing D8594972, covering a detail of Drawing F8594965.

CRITICAL	
1. Metal defective	Visual (Probe light)
MAJOR	
101. Pitch diameter of spike thread, max.	Gage
102. Minor diameter of spike thread, max.	Gage
103. Pitch diameter of chamber thread, max.	Gage
104. Minor diameter of chamber thread, max.	Gage
105. Diameter rear of cone seat	Gage
106. Diameter of "O" ring counterbore	Gage
107. Diameter of body	Gage
108. Diameter of bourrelet, min.	Gage
109. Diameter rear of spike thread	Gage

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4.3.2.4 (Cont'd)

	Defect	Method of Inspection
110.	Depth of "O" ring counterbore	Gage
111.	Depth from front face to cone seat	Gage
112.	Length to outside basic diameter	Gage
113.	Length to forward inside basic diameter	Gage
114.	Length to rear inside basic diameter	Gage
115.	Length from front face to rear of forward bourrelet	Gage
116.	Length from rear face to front of rear bourrelet	Gage
117.	Length of effective spike thread, min.	Gage
118.	Length from rear face to forward end of chamber thread	Gage
119.	Length, total	Gage
120.	Variation in wall thickness (indicated area)	Gage
121.	Angle of front face	Gage
MINOR		
201.	Diameter of spike thread counterbore	Gage
202.	Depth of spike thread counterbore	Gage
203.	Radii or chamfers missing or incorrect	Visual (See 6.1)
204.	Surface finish improper	Visual (See 6.1)
205.	Protective coating inadequate or damaged for indicated surface	Visual
206.	Burr	Visual
207.	Foreign matter	Visual

4.3.2.5 Chamber, Drawing F8594974, covering a detail of Drawing F8594965.

CRITICAL		
1.	Metal defective	Visual
MAJOR		
101.	Runout of obturating band diameter with pitch diameter of body thread and body seat	Gage
102.	Pitch diameter of body thread, min.	Gage
103.	Major diameter of body thread, min.	Gage
104.	Diameter of undercut rear of body thread	Gage
105.	Diameter of fuze cavity	Gage
106.	Diameter of obturating band seat	Gage
107.	Diameter of conduit hole	Gage

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4.3.2.5 (Cont'd)

	Defect	Method of Inspection
108.	Depth of fuze pin slot	Gage
109.	Depth from rear face to fuze seat	Gage
110.	Distance to bottom of fuze pin slot	Gage
111.	Length to basic diameter	Gage
112.	Length from front face to body seat	Gage
113.	Length from front face to large outside diameter	Gage
114.	Length, total	Gage
115.	Location of conduit hole	Gage

MINOR

201.	Diameter of boom thread undercut (if applicable)	Gage
202.	Width of boom thread undercut (if applicable)	Gage
203.	Width of body thread undercut	Gage
204.	Radii or chamfers missing or incorrect	Visual (See 6.1)
205.	Surface finish improper	Visual (See 6.1)
206.	Burr	Visual
207.	Foreign matter	Visual

4.3.2.6 Band, Obturator, Drawing D8594969, covering a detail of Drawing F8594965.

CRITICAL

None defined

MAJOR

101.	Diameter, inside	Gage
102.	Thickness of band	Gage
103.	Width of band	Gage
104.	Angle missing or incorrect	Visual (See 6.1)
105.	Grooves missing or incorrect	Visual (See 6.1)
106.	Excessive flash	Visual
107.	Material defective	Visual

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Direction arrow missing or incorrect	Visual (See 6.1)
202. Radii or chamfers missing or incorrect	Visual (See 6.1)
203. Surface finish improper	Visual (See 6.1)
204. Burr	Visual
205. Foreign matter	Visual

4.3.2.7 Plug, Fuze Lock, Drawing C8594966, covering a detail of Drawing F8594965.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Pitch diameter of thread, max.	Gage
103. Major diameter of thread, min.	Gage
104. Thickness	Gage
105. Metal defective	Visual
MINOR	
201. Presence and dimensions of wrench holes	Visual (See 6.1)
202. Presence and dimensions of recess	Visual (See 6.1)
203. Radii or chamfers missing or incorrect	Visual (See 6.1)
204. Surface finish improper	Visual (See 6.1)
205. Burr	Visual
206. Foreign matter	Visual

4.3.2.8 Boom, Drawing D8594973, covering a detail of Drawing D8594989.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of fin thread, min.	Gage
102. Major diameter of fin thread, min.	Gage
103. Length of fin thread pilot	Gage
104. Length from chamber shoulder to fin shoulder	Gage
105. Length from fin shoulder to rear of fin thread	Gage
106. Length, total, max.	Gage

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of fin thread undercut	Gage
202. Diameter of fin thread pilot	Gage
203. Width of fin thread undercut	Gage
204. Radii or chamfers missing or incorrect	Visual (See 6.1)
205. Surface finish improper	Visual (See 6.1)
206. Burr	Visual
207. Foreign matter	Visual

4.3.2.9 Fin, Drawing D8594976, covering a detail of Drawing D8594989.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. True position of hub with pitch diameter of boom thread and front face	Gage
102. True position of tracer cavity with minor diameter of rear thread	Gage
103. Parallelism of fin blades with pitch diameter of boom thread and front face	Gage
104. Pitch diameter of boom thread, max.	Gage
105. Minor diameter of boom thread, max.	Gage
106. Length of effective boom thread, min.	Gage
MINOR	
201. Depth of boom thread cavity	Gage
202. Length, total	Gage
203. Radii or chamfers missing or incorrect	Visual (See 6.1)
204. Surface finish improper	Visual (See 6.1)
205. Burr	Visual
206. Foreign matter	Visual

4.3.2.10 Fin and Boom Assembly, Drawing D8594989, covering a detail of Drawing F8594965.

CRITICAL	
1. Metal defective	Visual

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4.3.2.10 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Weight	Approved scale
102. Runout of boom taper with pitch diameter of chamber thread and chamber shoulder	Gage
103. Runout of small boom diameter with pitch diameter of chamber thread and chamber shoulder	Gage
104. Runout of outside diameter of fin blades with pitch diameter of chamber threads and chamber shoulder	Gage
105. Pitch diameter of chamber thread, min.	Gage
106. Pitch diameter of chamber thread, max.	Gage
107. Major diameter of chamber thread, min.	Gage
108. Pitch diameter of tracer thread, max.	Gage
109. Pitch diameter of tracer thread, min.	Gage
110. Minor diameter of tracer thread, max.	Gage
111. Diameter over fin blades	Gage
112. Diameter of fin hub	Gage
113. Diameter of tracer thread undercut, max. (if applicable)	Gage
114. Diameter of tracer cavity	Gage
115. Diameter rear of boom taper	Gage
116. Diameter of chamber shoulder	Gage
117. Diameter of chamber thread undercut	Gage
118. Depth from rear face to bottom of tracer cavity	Gage
119. Length of effective tracer thread, min.	Gage
120. Length from front face to chamber shoulder	Gage
121. Length, total, max.	Gage
122. Width of chamber thread undercut	Gage
123. Gap between fin and boom	Visual
MINOR	
201. Depth from rear face to start of tracer cavity, max.	Gage
202. Radii or chamfers missing or incorrect	Visual (See 6.1)
203. Surface finish improper	Visual (See 6.1)
204. Burr	Visual
205. Foreign matter	Visual

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4.3.2.11 Ring Locking, Drawing C8595232, covering a detail of Drawing F8594965.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Runout of inside diameter with pitch diameter of thread and rear face	Gage
102. Pitch diameter of thread, min.	Gage
103. Major diameter of thread, min.	Gage
104. Diameter of thread pilot	Gage
105. Diameter, inside	Gage
106. Length of thread pilot	Gage
107. Metal defective	Visual
MINOR	
201. Length, total	Gage
202. Presence and dimensions of wrench slots or holes (as applicable)	Visual (See 6.1)
203. Radii or chamfers missing or incorrect	Visual (See 6.1)
204. Surface finish improper	Visual (See 6.1)
205. Burr	Visual
206. Foreign matter	Visual

4.3.2.12 Projectile Sub-Assembly, Drawing F10542957, covering a detail of Drawing F8594965, (Prior to paint).

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Weight	Approved scale
102. Runout of pitch diameter of nose cap thread with bourrelets and nose cap seat	Gage
103. Runout of forward end of spike stem with bourrelets	Gage
104. Runout of spike flange with bourrelets	Gage
105. Runout of body diameter with bourrelets	Gage
106. Runout of chamber with bourrelets	Gage
107. Runout of pitch diameter of boom thread with bourrelets and rear face of chamber	Gage
108. Runout of cone with bourrelets	Gage

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4.3.2.12 (Cont'd)

	Defect	Method of Inspection
109.	Perpendicularity of front face of spike flange with bourrelets	Gage
110.	Diameter of spike flange	Gage
111.	Diameter of chamber large	Gage
112.	Diameter at rear face of chamber	Gage
113.	Diameter of "O" ring groove at rear face of chamber	Gage
114.	Depth of "O" ring groove at rear face of chamber	Gage
115.	Gap between body and spike	Gage
116.	Width of "O" ring groove at rear face of chamber	Gage
117.	Length from bottom of boom cavity to wire terminal	Visual (See 6.1)
118.	Length between wire terminal and conduit	Visual (See 6.1)
119.	Improper assembly	Visual and Manual
120.	Obturating band missing or damaged	Visual
121.	Direction of arrow on obturating band incorrect	Visual
122.	Burr	Visual
123.	Foreign matter	Visual

MINOR

201.	Length, total, max.	Gage
202.	Conduit hole in cone not properly aligned with chamber hole	Visual

4.3.2.13 Metal Parts Assembly, Drawing F8594965, (After painting).

CRITICAL

1.	Metal defective	Visual

MAJOR

101.	Pitch diameter of nose cap thread, min.	Gage
102.	Pitch diameter of nose cap thread, max.	Gage
103.	Major diameter of nose cap thread, min.	Gage
104.	Pitch diameter of boom thread, max.	Gage
105.	Pitch diameter of boom thread, min.	Gage
106.	Minor diameter of boom thread, max.	Gage
107.	Diameter of bourrelets, max.	Gage

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4.3.2.13 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Paint not dry	Visual/Tactile
202. Paint on threads (excessive)	Visual
203. Protective coating inadequate or damaged	Visual
204. Protector caps missing or damaged	Visual
205. Marking missing, incorrect or illegible	Visual

4.3.2.14 Plug and Disc Assembly, Drawing B8597489, covering a detail of Drawing F8594965.

CRITICAL	
None defined	
MAJOR	
101. True position of inside diameter with major diameter of thread	Gage
102. Pitch diameter of thread, max.	Gage
103. Pitch diameter of thread, min.	Gage
104. Major diameter of thread, min.	Gage
105. Thickness of disc	Gage
106. Thickness through body	Gage
107. Improper assembly	Visual/Manual
108. Metal defective	Visual
109. Foreign matter	Visual
110. Threads damaged or not full	Visual
MINOR	
201. Diameter of thread chamfer	Gage
202. Distance between wrench holes	Visual (See 6.1)
203. Surface finish improper	Visual (See 6.1)
204. Burr	Visual

4.3.2.15 Wire Assembly, Drawing C10521897, covering a detail of Drawing F8594965.

CRITICAL	
None defined	

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4.3.2.15 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter of hole in eyelet terminal	Gage
102. Diameter of insulator head (large)	Gage
103. Diameter of insulator head (small)	Gage
104. Length of insulator head (large)	Gage
105. Length, total	Gage
106. Components damaged or defective	Visual/Manual
107. Improper assembly	Visual/Manual
MINOR	
201. Excessive flash (insulator)	Visual
202. Burr	Visual
203. Foreign matter	Visual

4.3.2.16 Conduit, Drawing B8594968, covering a detail of Drawing F8594965.

CRITICAL

None defined

MAJOR	
101. Thickness of wall, min.	Gage
102. Material defective	Visual
MINOR	

None defined

4.3.2.17 Pallet, Drawing F8597604 for Metal Parts Assemblies.

CRITICAL

None defined

MAJOR

None defined

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4.3.2.17 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Strapping missing, loose or broken	Visual/Manual
202. Pallet broken	Visual
203. Improperly packed	Visual

4.3.2.18 Cartons, prior to sealing.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Support missing or damaged	Visual
202. Separator missing or damaged	Visual
203. Improperly packed	Visual
204. Incorrect number in carton	Visual
After sealing.	
205. Contents loose	Manual
206. Carton damaged, cut or gouged through all layers	Visual
207. Carton not completely sealed	Visual
208. Marking missing, incorrect or illegible	Visual

5. (U) PREPARATION FOR DELIVERY

5.1 (U) Preservation and packaging.-Not applicable.

5.2 (U) Packing.

5.2.1 (U) Level A - Military pack.-Not applicable.

5.2.2 (U) Level B - Limited military pack.-Not applicable.

5.2.3 (U) Level C - Minimum military pack.-The metal parts for projectile, HEAT-FS, 90MM, M431 shall be packed as follows:

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5.2.3.1 (U) Metal Parts Assembly, F8594965.-Shall be packed in accordance with Drawing F7548604.

5.2.3.2 (U) Fin and Boom Assembly, D8594989.-Shall be packed in accordance with MIL-STD-1169, pack 40 fin and boom assemblies in 2 layers in a 4 x 5 pattern using interlocking separators.

5.2.3.3 (U) Fuze Lock Plug, B8594966.-Shall be packed in accordance with MIL-STD-1169, pack 500 fuze lock plugs in 2 layers arranged in a 10 x 5 pattern, 5 to a cell, using interlocking separators.

5.2.3.4 (U) Seal "O" Ring, MS9021-227.-Shall be packed in bulk in accordance with MIL-P-10025, in quantities not to exceed 2500.

5.2.3.5 (U) Plug and Disc Assembly, B8597489.-Shall be packed in accordance with MIL-STD-1169. Pack 2000 plug and disc assemblies in 2 layers arranged in a 10 x 10 pattern, 10 to a cell, using interlocking separators.

5.2.3.6 (U) Percentage of extra parts packing.-Packing for the required percentage of extra parts shall be provided for by furnishing the proper number of prime containers.

5.2.3.7 (U) Shipping.-When component parts from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots clearly indicated.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of a visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards.-Visual standards will be established at contractors plant in compliance with 3.8.3 or whenever directed by the Procuring Contracting Agency.

6.3 Alternative testing procedures.-Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to, or better than, the quality assurance provisions specified herein.

6.4 When a product is being inspected in accordance with MIL-STD-1235 and a defective characteristic is found, product produced prior to the discovered defect will be screened until the frequency (f) is found to be free of that defect.

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6.5 Fuze malfunction is not considered as failure to comply with 3.8.

6.6 Changes in form, size or temper of the flash disc, and significant changes in feeds and speeds of the flo-turn operation are considered a major change in process.

6.7 Ordering data.-Procurement documents should specify the title, number and date of this specification.

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